

# Canals - a canal-building mod for Civilization VII

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# Canals

A Civilization VII mod. Build a Canal on a narrow neck of land between two bodies of water. When it is finished the tile becomes water, drawn as a canal for its age, and ships sail straight through it. It stays a canal across every save and reload, and the AI digs canals and sails through them too.



Figure 1: A two-tile Exploration canal at Roma, after a save and reload

*A two-tile Exploration cut at Roma, joining the northern sea to the city's estuary, after a save and reload. The channel, the quays and the moored boats are drawn by the mod; the canal is still water, and still a canal.*

The Canal is an ordinary building in the production list, and a town buys it with gold like any other building. Each age has its own Canal, unlocked by one of that age's techs, and a settlement can dig as many canals as it has sites for. The mod decides where a Canal may go, turns the tile into water when it completes, and dresses it. Everything else, the ship pathing, the tile's ownership, the citizen who works it and the yields it gives, is the game's own state.

## What the player does

Pick the Canal in a city's production list, or buy it with gold in a town's purchase list (or a city's). It is a building; each age has its own, unlocked by a tech, and a settlement can have any number of them, one per site:

| Age         | Building       | Unlocked by       | Production cost |
|-------------|----------------|-------------------|-----------------|
| Antiquity   | Ancient Canal  | Engineering       | 300             |
| Exploration | Medieval Canal | Shipbuilding      | 500             |
| Modern      | Modern Canal   | Industrialization | 800             |

A bought Canal is finished at once: the tile turns to water as soon as the purchase goes through.

On the placement screen the city offers only the tiles a canal makes sense on. In Antiquity that is a land tile the city owns with water on at least two separate sides: a strip of land between two shores. Sea, lake and navigable river all count, so a canal can join a river to the sea or two reaches of a river. Three coast tiles on one side and one on the other qualifies; a headland with all its water in one stretch does not.



Figure 2: The neck of land at Roma before the cut

*The site before the first Canal is ordered: the neck of land between Roma's estuary and the northern sea.*

Choose a tile. It becomes an urban district holding the Canal under construction, and the city builds it like any other building.

When the Canal completes, the tile turns to water and is drawn as a canal at once, dressed for the age it was dug in: a water channel cut through the land between the shores it joins, boats moored where it meets open water, and along the banks an Antiquity harbor with its stone quay, or Exploration houses and a wharf lining the channel, or Modern buildings and a harbor. Any ship can now path through the tile between the two waters, in the same turn. The channel runs between two shores rather than toward every side the tile has water on, so a tile standing in open water is not drawn as a star. Where the canal touches another body of water as well, a lake beside the cut for instance, one branch opens into it, from whichever tile of the canal suits it best, so it is plain that ships can come in that way too.



Figure 3: A Cog in the finished two-tile cut

*The finished cut with a Cog sailing through. Each Canal that opens beside an earlier one redraws both, so the channels meet and the cut reads as one waterway.*

## Longer canals

From the Exploration age a canal can be two tiles long: a second Canal is offered on the tile beside the first that continues it in a straight line, and nowhere else around it. That is the whole Exploration allowance: no third tile, no bend, no branch. A queued Canal counts as part of the run, so the second can be ordered while the first is still being dug; ships pass once both are finished.

A canal can cross from one settlement's land into another's: each tile is built or bought by the city or town that owns it, and a tile beside another settlement's canal is offered as the next tile of that canal. City and town, two towns, or two cities make one waterway alike.

In the Modern age a run may be up to five tiles long and may bend and branch, as long as every tile of it reaches open water through the run. A junction where a branch leaves the main cut is drawn as an open basin, with no houses on it.

| Age         | Run                              | Bends and branches |
|-------------|----------------------------------|--------------------|
| Antiquity   | 1 tile                           | no                 |
| Exploration | a straight line of up to 2 tiles | no                 |
| Modern      | up to 5 tiles                    | yes                |

In Antiquity a canal is always a single tile: the tiles beside an existing or queued canal are not offered. In every age a tile beside a canal is judged as part of that canal's run; a canal's own water never qualifies a neighboring tile as an isthmus by itself.



Figure 4: An Antiquity canal from the city center to the sea, with a Galley in it

*An Antiquity canal dug from the city's own tile to the sea, with a Galley in the channel. A settlement center counts as a canal end, so a city or town can open its center to the water.*



Figure 5: The same Antiquity canal after a save and reload

*The same canal after a save and reload: the quay, the boats and the channel through the land are all still there.*

## Which tiles qualify

- flat or hill land the city owns, within the city's build ring, not a navigable river tile;
- at least two separate ways in for a ship: sea, lake or navigable river on separate sides of the hex, in any combination (the same sea on both sides counts), ice not counted; two different bodies of water around it also qualify, even where they meet in one stretch; a settlement's own center beside the tile is a way in of its own, so a city or town can dig a canal from its center out to the sea or a river, and a longer run may start or end at the center. A center counts apart from the water rather than as a piece of it, so a settlement standing next to a neck never costs the neck its two shores;
- from Exploration on, alternatively the tile that extends an existing or queued canal within the age's run rule: a straight two-tile line in Exploration, up to five tiles with bends and branches in Modern;
- no urban district on it, and no building other than a rural improvement, which is removed when the Canal is placed;
- no unit standing on it;
- no resource on it: a tile holding a resource can take no building, and the Canal is one;
- not already a canal, dug or queued.

## What it gives, and what it costs

The Canal, like other buildings, takes one citizen to build. When it completes that citizen moves onto the finished canal, which becomes a worked fishing tile of the city, and the Canal building stays on it as an ordinary building of the city with its own yields, by the age it was dug in. All of it is ordinary city yield, counted where the city counts everything else. If the city cannot place the citizen on the canal itself, the tile still becomes a rural tile of the city and the citizen is left for you to place.

| Canal       | Food | Gold | Ships through it per turn |
|-------------|------|------|---------------------------|
| Antiquity   | 2    | 2    | 1                         |
| Exploration | 3    | 4    | 2                         |
| Modern      | 4    | 6    | no limit                  |

The ship limit holds your own move orders: a move whose path runs through a canal that has already taken its ships this turn is not sent, and the count starts again each turn. A ship already sitting in the canal is not counted. The game's AI plots its routes natively and is not held.

The cost is the production and the tile: the rural improvement on it, if any, is gone, and the tile is water from then on.



Figure 6: A branched Modern canal system at Roma, with an Ironclad in the channel

*A Modern run of five Canals at Roma that bends and branches between the northern sea, the estuary and the river, with an Ironclad in the channel and the age's harbors and buildings along the banks.*

## Cliffs and locks

A canal can be cut through a tile with cliffs on its edges. The cliffs stop mattering the moment the canal opens: ships cross every edge of the tile. The rock faces themselves stay in the picture until the game is next loaded, so where a channel meets a cliff shore the canal is drawn stepping down through a lock: a stone chamber with gates across the channel, a gatehouse on one bank, a water wheel on the other, and water spilling over the cliff foot beyond the gates. This is cosmetic only.

## AI canals



Figure 7: A canal dug by an AI player across a snowy neck of land

*A canal an AI player dug on its own land, joining a lake to the sea across a snowy neck.*



Figure 8: An AI canal running through a town from the river to the harbor

*Another AI canal, cut through the houses of a town from the river down to its harbor.*

An AI player with the Canal unlocked digs one on a tile it works that joins two separate bodies of water. Its city builds the Canal with its production, or buys it outright when it has gold to spare, and the AI's units sail through it. Each AI waits eight turns between canals.

## Compatibility

- Adds three buildings and their tech unlocks, changes no base-game rows, replaces no base-game files.
- A save loads with the mod on or off. Saves hold each canal as the land it was cut from, and the mod turns it back to water when the game is loaded with the mod on. Without the mod a dug canal is a plain land tile of the city: its fishing boat and district stay, the Canal building and its yields go, and a ship left in the canal is stranded on the land. Move your ships out of the canals before you remove the mod.
- Added to a game already under way, after the Canal's tech was researched, the game keeps that age's Canal locked. The mod then sells it for gold in each settlement's purchase list instead of production, and says so once.
- While a canal is open the mod writes the autosave itself, at the start of your turn, on your own frequency and keep count. The files carry on the game's own series (AutoSave\_01\_0072 and onward), so the newest is at the top of the Autosaves tab and is the one Continue loads. The game's own autosave is held meanwhile; your setting is put back whenever the main menu loads.
- A save from Canals 1.0.0 holds its canals as open water, so they load drawn as open sea. The mod offers to reload at once, which draws them as canals; otherwise they look right after your next save and load.
- Single-player games only. In a multiplayer game no Canal can be built, and the mod says so when the game starts.
- English only for now. The mod's text is a handful of strings.

## Installation

1. Subscribe on the Steam Workshop, or download the zip from the latest release.
2. Unzip it so the **canals** folder sits in the Civilization VII Mods directory.
3. Enable **Canals** from Additional Content in-game.

## How it works

Five steps, each watched in the game on 2026-09-25, the purchase path on 2026-09-27 (Civilization VII 1.5.0). The run that proved each one is in docs/verification-runs.md.

1. **Placement.** The Canal buildings carry no terrain rule of their own. The mod wraps the engine's placement queries, for production and for purchase with gold (a town's only way to get a building): for a Canal, the offered plots are the settlement's land tiles within its build ring that pass the rule above, and the per-plot check refuses anything else with the mod's own message. The engine's verdict on *whether* a Canal may be had (locked, already queued, not enough gold) is kept; the mod decides only *where*.
2. **Commit.** The engine offers buildings only on tiles that already hold an urban district, so a Canal on a rural isthmus first gets one, created by script, after which the Canal is offered there and a real build order queues it, or a real purchase buys it. A bought Canal arrives complete and the engine sends no completion event for it, so the mod opens it straight away. The plot is remembered in the save, so a reload mid-construction still finishes the job. If the engine does not take the order within a few seconds, the district is removed again and the plot bought back.
3. **Completion.** When the Canal completes, the handler retypes the tile to coast (the feature cleared first), removes the Canal and its urban district (a district's housing block would hide the canal), buys the released plot back for the city, frees the Canal's citizen as a pending point and places it on the new water tile with the city's own expand order (a rural district with fishing boats, worked by that citizen), and re-creates the Canal building on that rural district, where its yield rows count in the city at once.
4. **The passage.** A tile retyped to coast is pathable by ships immediately: a ship's path to the far side goes from around the land to through the tile in the same turn. The tile's area id does not change until the age transition, which is why the placement rule resolves a canal tile's waters through its neighbors rather than its area.
5. **The look.** A retyped hex keeps its land mesh, and a load draws it as land too (step 6), so the canal is drawn by script from shipped meshes: river channel pieces meeting at the hex center - one to each canal beside it, the two shores that lie most nearly opposite, and one branch into each other body of water the canal touches - the age's houses and harbor along the first arm, moored boats where the canal meets open water or a junction, and lock pieces at any shore that was a cliff. Three looks per age, chosen by plot number so a canal keeps its look across reloads. The opened canals are kept in the save and redrawn on load and at the start of every turn.
6. **AI canals.** At the start of each of your turns an AI with the Canal unlocked looks for a tile it works that joins two separate bodies of water. When it finds one, the city builds a Canal there with its production, as yours do, or buys it outright if it has the gold with a fifth to spare. The canal then opens as yours do, and AI units sail through it. Each AI waits eight turns between canals.
7. **Saving.** A load draws each hex from the terrain in the save, and a coast hex comes out as open sea, so every save holds the canals as land. The terrain must change before a save starts, never during one: a retype while the game is writing a save crashes it. Every save goes through the game's save call, which the mod wraps: the canals turn to land, the save is sent once the map reads them as land, and they are coast again when it completes. The game's autosave comes too soon after the AI turns for that, so the mod writes the autosave itself at the start of your turn; the canals are water through every AI turn. After a load they stay land until the game has started and the save the engine makes on loading an autosave is done, then turn coast; the land mesh the load drew stays in place under the mod's channel. When an age ends the canals turn to land and stay land, so the next age, which is built from the map the old one leaves, loads them as land like any save.

A safety sweep on load and at the start of every turn re-checks every remembered site and every complete Canal on the map, so a missed completion event is caught on the next turn. A completed Canal that shares its tile with other buildings (an AI's, in an urban slot) or does not sit on a canal site is left alone as a building.

## Status

Watched working end to end on Civilization VII 1.5.0, 2026-09-25, in games of all three ages: placement through the real production screen, the district and the build order, the completion handler opening the canal, the citizen placed and the yields counted in the city, a Galley, a Cog and an Ironclad each sailing through, the three-tile Exploration cut and the Modern trunk ordered in sequence through the real build path, the lock dressing at a cliff shore, and short AI soaks with the Canal unlocked for every player without a crash. On 2026-09-27 a town bought a Medieval Canal through the purchase path: the placement screen offered the canal site alone, the canal opened at once, the gold was charged once and a Cog sailed through. On 2026-09-28 each age's Canal was watched in the lists a player reads it from, named and with its own icon: the city's production list, the city's purchase tab and a town's purchase list, in games of all three ages. Also on 2026-09-28, on a canal opened through the real build path: the autosave, a script save and

a quicksave through the game's own save model each wrote the canal as land, with no crash; the script save and the mod's own autosave each reloaded with the canal drawn through land and a ship pathing through it, and the canal was water through every AI turn. AI players started canals on their own in autoplaid games, paid for them with their cities' production and opened them; two AI canals placed by the planner opened for their owners on a turn-66 map, and the AI's embarked units moved through one on three turns. A two-tile cut held by two cities, one built with production and one bought, still carried a ship through on the release build. An age transition with the mod running went through without a crash, and on 2026-09-28 one with a canal open: the canal crossed into the Modern age as land, kept its record, and was drawn as a canal through the land once the new age loaded. Five Modern canals dug through the real build path reloaded drawn as canals. A save holding a canal as water (from Canals 1.0.0) was offered a reload, and after it the canal was drawn through land. The mod's autosaves carried on the game's own series and were what the load list and Continue showed first. In a game the mod was added to after Engineering, a town bought the Canal from the mod for gold, charged once, and it opened.

Not watched:

- A refusal by the per-turn ship limit.
- A Modern branch built through the real build path: the branch tile was offered, but the engine dropped that one build order with no reason code in the gallery run, and branching was otherwise proved by retyping the tiles directly.
- The second and third look of each age as whole compositions (their pieces were photographed individually).
- An AI warship crossing between two seas through a canal: the AI units seen in canals were embarked land units.
- The age transition through the game's own end-of-age screens: the watched one ran under Autoplay.
- A ship's path through a canal after an age transition.
- An AI digging a Canal whose tech it researched before the mod was added.

Known limits:

- Multiplayer: the retype is a local call, so the mod does not offer the Canal in a network game.
- A save from Canals 1.0.0 draws its canals as open sea until it is saved and loaded once more.

## Layout

|                                     |                                                                                                   |
|-------------------------------------|---------------------------------------------------------------------------------------------------|
| canals.modinfo                      | data and one script                                                                               |
| data/canals.xml                     | the three Canal buildings: cost, citizen, yields, districts                                       |
| data/canals-antiquity.xml           | each age's tech unlock, loaded only while that age is in use                                      |
| data/canals-exploration.xml         |                                                                                                   |
| data/canals-modern.xml              |                                                                                                   |
| ui/canals.js                        | the mod: placement, commit, completion, looks, locks, ship limit, saving, AI canals, safety sweep |
| ui/canals-shell.js                  | main menu: puts the player's autosave frequency back                                              |
| text/en_us/CanalsText.xml           | name, description and the placement message                                                       |
| docs/verification-runs.md           | every step and the run that proved it                                                             |
| docs/steam-workshop-description.txt | the Workshop page text                                                                            |
| docs/workshop-preview.png           | the mod icon and Workshop preview (rendered from workshop-preview.svg)                            |
| gallery/                            | the screenshots used above                                                                        |
| scripts/build_readme_pdf.sh         | builds README.pdf from this file (npm run readme:pdf)                                             |
| scripts/steam-changelog.mjs         | renders CHANGELOG.md into CHANGELOG.steam.txt for the Workshop                                    |
| release.sh                          | builds the release zip and the Workshop manifest into dist/                                       |
| CHANGELOG.md                        | what changed in each version                                                                      |
| README.pdf                          | this file, typeset                                                                                |

In a running game, `globalThis.__canals.enabled = false` makes every hook pass straight through, and `globalThis.__canals.uninstall()` restores the engine's own methods; both are for troubleshooting.

The routes that were tried and set aside, the probes and the harness runs live in the author's working tree (`mod_ideas_tested/canals`), outside this folder.

## Credits

Built by Tower.